

1. [5 pts] Find the general antiderivative of the following functions. Use C as an arbitrary constant. and use $F(x)$ as the general antiderivative of $f(x)$.

1). $f(x) = 2 \cos x$

2). $f(x) = \frac{1}{x^3}$

2. [5 pts] Find the following indefinite integrals. Use C as an arbitrary constant.

1). $\int \frac{1}{1+x^2} dx$

2). $\int \frac{x^4 + \sqrt{x}}{x} dx$